

Evaluating the attribution of bivalve-macroalgae polyculture as a carbon source or sink from an ecosystem perspective: A case study of kelp-oyster IMTA

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ABSTRACT

To clarify interactions among bivalves, macroalgae, and water column in the bivalve-macroalgae polyculture system, in-situ mesocosm cultivation was conducted for 24 h in both summer and winter. These experiments assessed seasonal variations in CO₂ and components of dissolved inorganic carbon (DIC), nitrogen (DIN), and phosphorus (DIP) under three modes: monoculture of three sizes of oysters (*Crassostrea gigas*) (OA group), monoculture of kelp (*Saccharina latissima*) (KA group), and polyculture of oysters and kelp (KO group) at fixed wet weight ratio of 800: 200 g. The results revealed the OA group functioned as a CO₂ source in both summer and winter, with summer increase rates of DIC, DIN, and DIP by oysters much higher than in winter and decreasing with larger oyster size. Pearson correlation analysis indicated a negative correlation between soft tissue dry weight of oysters and concentrations of NO₂⁻-N and NH₄⁺-N. While, the KA group functioned as a CO₂ sink, with winter decrease rates of DIC, DIN, and DIP by kelp being 14.24, 3.16, and 5.54 times greater than in summer, respectively. In the KO groups, oyster size and seasonal factors significantly affected water parameters, including pH, dissolved oxygen (DO), pCO₂, and various DIC, DIN, and DIP components ($P < 0.05$) and the system released DIC, DIN, and DIP in summer while absorbing them in winter. Subtraction calculations indicated that the culturing water in KO groups absorbed CO₂, DIC, DIN, and DIP during both seasons, suggesting a synergistic bioremediation effect between phytoplankton and macroalgae at an appropriate cultivation ratio.

1. Introduction

Aquaculture has emerged as one of the fastest-growing sectors in global food production in recent years, significantly contributing to food security and nutrition worldwide (FAO, 2024). However, it has also led to a series of environmental challenges, including excessive greenhouse gas emissions and increased risks of eutrophication, which hinder its sustainable development (Wang et al., 2015; Zhang et al., 2022). Integrated Multi-Trophic Aquaculture (IMTA) systems, which involve the co-cultivation of species occupying different ecological niches and trophic levels, can enhance nutrient recovery rates and promote carbon sequestration. This approach not only produces high-quality, nutrient-

rich biomass but also effectively mitigates the negative environmental impacts of aquaculture (Shahar and Guttman, 2020; Reid et al., 2020). The cultivation of unfed species, such as macroalgae and bivalves with the characteristics of low production costs, high resource utilization efficiency, complementary functions, and positive ecological roles in nutrient and carbon removal from aquatic ecosystems, which make them ideal candidates for IMTA systems (MacDonald et al., 2011; Marinho-Soriano et al., 2011). Currently, the co-cultivation of bivalves and macroalgae has been proven to be an effective aquaculture mode for alleviating environmental issues such as eutrophication and climate change associated with greenhouse gas emissions, and promoting the sustainable development of aquaculture (Fang et al., 2020; Jiang and

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Fang, 2021; Zhang et al., 2017).

The nutrient cycling and carbon balance in bivalve-macroalgae co-culture systems are influenced by the complex and ongoing interactions between the culturing water (including the planktonic plankton), bivalves, and macroalgae, as well as environmental changes. In the co-culture systems, the culturing water functions as a carbon sink when photosynthesis by phytoplankton (contributing over 50 % to global marine primary productivity) and photosynthetic bacteria predominates; when heterotrophic microbial respiration prevails, it becomes a carbon source (Wallace et al., 2014; Xiong et al., 2024). The ecosystem services provided by culturing water are jointly regulated by bivalves and macroalgae. Studies reveal that the biomass of planktonic organisms, such as phytoplankton, zooplankton, and bacteria, is influenced by top-down control exerted by the feeding activities of bivalves (Fournier et al., 2012; Simon et al., 2009) and the positive effects of localized nutrient enrichment from bivalve excretion, which primarily benefits phytoplankton (Trottet et al., 2008). Simultaneously, phytoplankton and heterotrophic microbial biomass are influenced by bottom-up processes driven by macroalgae, which regulate the availability of nutrients, light, and dissolved organic carbon. (Prins et al., 1995). In addition, the physiological and ecological functions of bivalves and macroalgae, as well as their interdependent interactions, profoundly influence the biogeochemical cycling of carbon, nitrogen, and phosphorus within the system. Bivalves transform a portion of the carbon, nitrogen, and phosphorus filtered from the environment into their soft tissues and shells (van der Schatte Olivier et al., 2020), while releasing residual nutrients back into the ecosystem in the form of CO₂, dissolved inorganic nitrogen (DIN), and dissolved inorganic phosphorus (DIP) during respiration, calcification, and excretion processes (Sea et al., 2022). These released nutrients provide essential inorganic carbon and nutrients to support the photosynthesis and growth of macroalgae and plankton within the water column. Macroalgae, in turn, play a pivotal role in mitigating excess nutrient accumulation by elevating pH and oxygen levels, improving the carbonate balance, and mitigating the accumulation of excess nutrients through photosynthetic carbon fixation and nutrient uptake, thereby creating favorable conditions for the growth and calcification of bivalves (Koweeck et al., 2017). Moreover, environmental variability induced by climate change—such as fluctuations in water temperature, CO₂ concentrations, and nutrient dynamics—can significantly alter the growth and metabolic activity of bivalves, macroalgae, and planktonic communities, thereby influencing the overall ecological performance of the co-culture system (Sivertsen and Bjørge, 2015; Zhao et al., 2011). These findings underscore the critical role of environmental factors and complex biological interactions in biogeochemical cycles and energy flow within these co-culture ecosystems.

Although previous studies have provided important insights into the cycling of carbon, nitrogen, and phosphorus in bivalve-macroalgae co-culture systems, there is still a lack of precise quantitative evaluations of the internal interactions among different components of these systems (including bivalves, macroalgae, and the water column), which are crucial for determining the ecological service functions of IMTA systems (Clemente and Thomsen, 2023). Furthermore, these interactions persist throughout the entire cultivation cycle of the farmed organisms and become increasingly complex as cultured organisms grow (Fang et al., 2018; Wang et al., 2005) and cultivation conditions change (Wang and Li, 2020). Therefore, there is a need for scientific methods to accurately and dynamically quantify these interactions. Sangou Bay, is a typical region for bivalve-macroalgae IMTA mode in northern China, where the longline aquaculture method is often practiced, resulting in relatively limited water exchange capabilities and leading to more pronounced interactions among different components of the aquaculture system in localized waters (Liu et al., 2022; Sun et al., 2020). Consequently, in situ co-culture experiments with kelp (*Saccharina japonica*) and oysters (*Crassostrea gigas*) of different sizes were conducted during both summer and winter in a semi-enclosed bay. We hypothesize that seasonal

changes and oyster size will alter the dynamic interactions between oysters, kelp, and the water column, thereby affecting the nutrient fluxes and carbon cycling in the system. By calculating the inorganic carbon and nutrient assimilate/release efficiencies of kelp and oysters in monoculture systems, and further employing a subtraction method to isolate the contributions of oysters and kelp from the total absorption/release values in co-culture system, thereby accurately quantifying the role of the water column, and the precise assessment of the dynamic changes in inorganic carbon and nutrient fluxes between the different components of the co-culture system were further obtained. In sum, this study proposes an alternative approach for comprehensively understanding how interactions among components within co-culture systems shape ecosystem services. Additionally, it provides valuable data support for refining management strategies of co-culture systems, maximizing resource utilization efficiency, and enhancing the environmental sustainability of these systems.

2. Materials and methods

2.1. Experimental design and management

In this study, an in situ cultivation experiment was conducted in Sanggou Bay (37°02'25"N, 122°33'29"E) (Fig. 1), China, in 2023, the small, medium, and large sizes of oysters (from different growth stages and with significant differences in wet body weight, Table S1) and kelp (with clean surfaces) collected from this sea area were selected to serve as experimental subjects. There are seven treatments were settled (Fig. 1, Table S2): kelp co-cultured with the small (KO_S), medium (KO_M), and large sizes of oyster (KO_L) treatment using a fixed wet weight ratio of oyster to kelp based on our previous findings (Fang et al., 2020); the small (OA_S), medium (OA_M), and large sizes of oyster monoculture (OA_L) treatment; the kelp monoculture (KA) treatment.

Specifically, the KO groups were cultured in an open enclosure device consisting of the 0.12 mm thick transparent polyethylene bag, plastic ring discs, and floating frame (10 × 10 m). In this device, 16 small-sized, 9 medium-sized, and 6 large-sized oysters, each size having a total wet weight of 800 g, were placed alongside 200 g of wet kelp fixed on a plastic ring disc in a 4:1 wet weight ratio and were suspended cultured in the transparent polyethylene bag filled with 500 L seawater (surface seawater, which was collected locally to ensure ecological relevance), which was secured to a floating frame (Fang et al., 2020). This system was selected to simulate natural, open-water conditions while allowing for controlled interaction among the oysters, kelp, and plankton. In contrast, the OA group was cultured in a sealed, airless enclosure (dissolved oxygen ≥5 mg/L at the end of the experiment), each enclosure contained one oyster (three sizes specified same as KO group) and a volume of filtered seawater (filtered by a 0.22 μm sterile filter). The oyster wet weight to water volume ratio was maintained at 8:5 (same as the KO group) to ensure comparability. Similarly, the KA group was also cultured in airless, sealed glass bottles (2.5 L), each containing 1 g of kelp and 2.5 L of filtered seawater, following the kelp wet weight to water volume ratio of 2:5 (referenced from the KO group) to ensure comparability. These two monoculture systems (OA and KO groups) were designed to calculate the inorganic carbon and nutrient assimilate/release rates of kelp and oysters per unit time (d) and unit wet weight (g). Each treatment group was suspended in similar water depths, with the culturing environment consistent with the natural seawater conditions concerning light exposure, temperature, and current disturbances. Five replications were settled in each treatment, and the pure seawater with no creatures served as the blank control (including KO_{non}, KA_{non}, and OA_{non} groups). The experiment was conducted for 24 h each in the summer and winter, starting from 17:00 on June 21 to 17:00 on June 22, and spanned from 19:00 on December 21 to 19:00 on December 22, guided by the local sunrise and sunset times. The salinity (S), temperature (T), dissolved oxygen (DO), and pH were monitored every 12 h in the KO group, and 500 mL water samples were

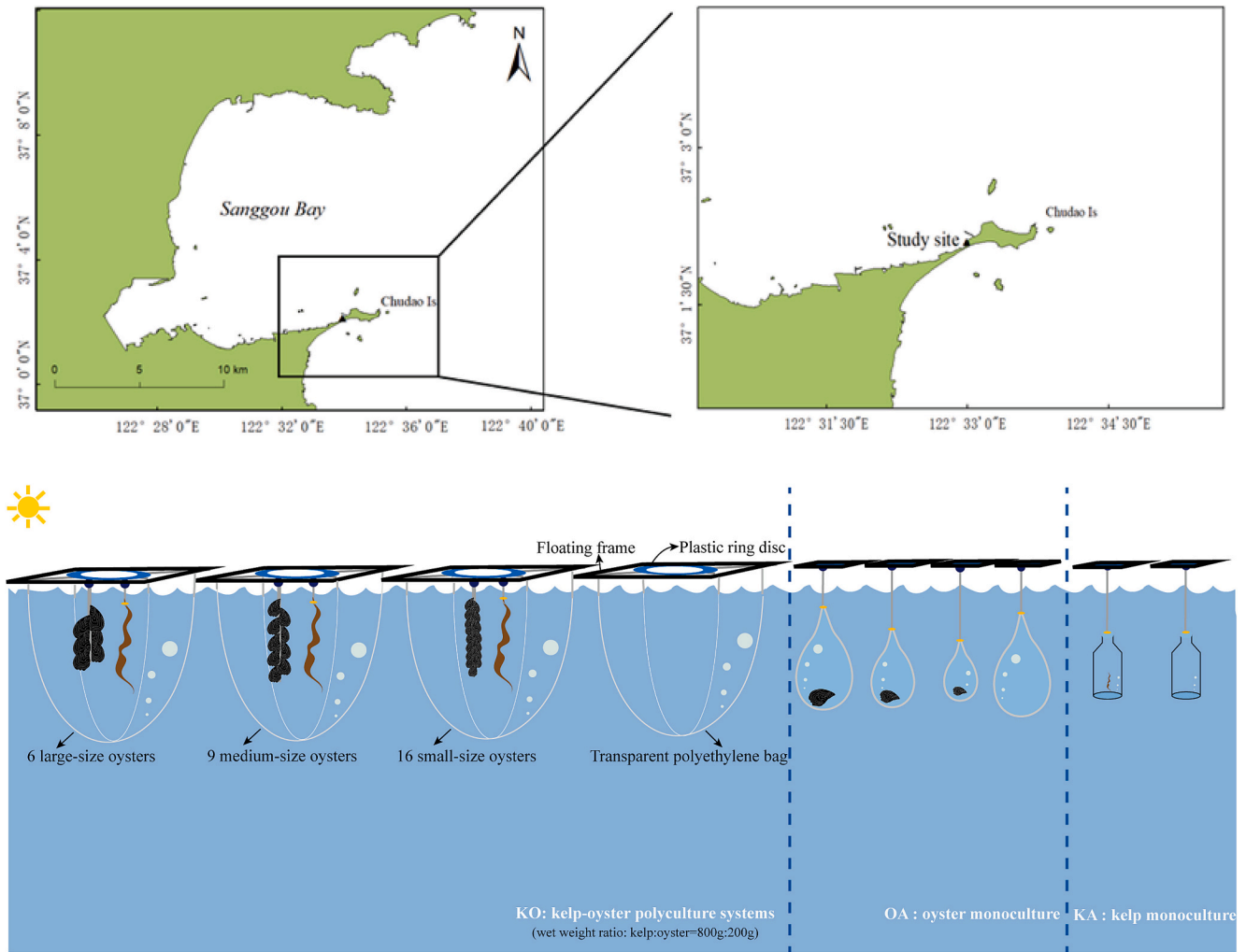


Fig. 1. The study site and experimental design.

collected from each treatment after the experiment, filtered (Whatman GF/F membranes) and fixed with HgCl_2 and CH_4Cl_3 , respectively, to analyze the components of the carbonate system (CO_2 , CO_3^{2-} , HCO_3^-) and nutrients.

2.2. Sample analysis

The S, T, DO, and pH of seawater were measured using a water quality analyzer (Eureka, USA), the pH measurement probe was calibrated daily according to the U.S. National Bureau of Standards (NBS) scale to ensure accurate measurements of pH_{NBS} (with a precision of ± 0.001). A total alkalinity titrator mode (AS-ALK2, Apollo SciTech, USA) was used to determine total alkalinity (TA). The parameters of the carbonate system, including total DIC, the partial pressure of CO_2 (pCO_2), and the saturation states of calcite (Ω_{cal}) and aragonite (Ω_{arag}) were further calculated by CO2SYS 2.1 software based on the data of pH_{NBS} values, TA, S, and T, the first and second dissociation constants for carbonic acid in seawater provided by Millero (2010) was utilized in the calculated process. The previous methods were referenced to determine the concentration of active phosphates ($\text{PO}_4^{3-}\text{-P}$), nitrite ($\text{NO}_2\text{-N}$), nitrate ($\text{NO}_3\text{-N}$), and ammonium ($\text{NH}_4\text{-N}$) (Murphy and Riley, 1962; Xie et al., 2023). The concentration of DIP was considered to be equal to the $\text{PO}_4^{3-}\text{-P}$. The amount of DIN and DIC were calculated as the sum concentrations of $\text{NO}_2\text{-N}$, $\text{NO}_3\text{-N}$, $\text{NH}_4\text{-N}$ and the total amount of CO_2 , CO_3^{2-} , HCO_3^- , respectively. The dry weight of tissue and shell was measured after the oyster was dried for 48 h at 70 °C.

2.3. Results calculation and statistics analysis

The amount of DIC, different components of the carbonate system, $\text{PO}_4^{3-}\text{-P}$, and different DIN forms increased (IA) or decreased (DA) by 1 oyster (with three sizes) or 1 g of kelp in monoculture systems (KA and OA groups) and by the culturing water (500 L) in the co-culture system (KO group) was calculated based on Eqs. (1) and (2), respectively. For comparison of the ability of aforementioned parameters decreased/increased by kelp and different sizes of oyster in two seasons, the biomass-specific increase/decrease rate was calculated by dividing the parameter value by the cultivation time (d) and the wet weight of kelp and oyster (g).

$$\text{IA} = [(C_0 - C_t) + (C_t' - C_0')] \times V. \quad (1)$$

$$\text{DA} = [(C_t - C_0) + (C_0' - C_t')] \times V.$$

$$C_{\text{KO-W}} = C_{\text{KO}} - C_{\text{OA}} \times M_0 - C_{\text{KA}} \times M_K. \quad (2)$$

In Eq. (1), C_0 and C_0' represent the initial concentrations of different components of the carbonate system and DIC in the treatment and control groups respectively, while C_t and C_t' represent their concentrations in these groups at the end of the experiment; the terms V in both equations correspond to the volume of the seawater in each monoculture system. In Eq. (2), C_{KO} , C_{OA} , C_{KA} , and $C_{\text{KO-W}}$ represent the amount of CO_2 , DIC, DIP, and DIN absorbed or released by the KO, OA, KA groups, and culturing water of the KO group; M_0 and M_K represent the number

of oysters and the wet weight of kelp in the KO group.

The experimental data were processed by the software IBM SPSS 26.0 (IBM Corporation, New York, USA). Shapiro-Wilk test and Levene's tests were conducted to test the normality and homogeneity of variables. Two-way ANOVA was conducted to test the statistically significant influence on physicochemical parameters of the water column in season, oyster's size, and their interaction. One-way analysis of variance (ANOVA) combined with Tukey's test was conducted for differences in water column indicators under different culturing modes in the same season. Student's *t*-test was conducted for seasonal differences of water column indicators under one culturing mode. Pearson's correlation analysis was conducted to examine the relationship between different water column indicators. The significance level was set at $P < 0.05$ to determine the significant differences and correlations of statistical data. All the experimental data were performed as mean values $\pm$ standard deviation (S.D.) and further visualized by software Origin 2023.

3. Results

The temperature and salinity differences between treatment groups at the end of the experiment were relatively small in both summer and winter: in summer, temperatures ranged from 25.06 to 25.26 °C and salinity from 30.79 to 30.81; in winter, temperatures ranged from 1.92 to 2.12 °C and salinity from 29.6 to 29.66.

3.1. Changes in physicochemical parameters of the water column in kelp-oyster polyculture mode

The Two-Way ANOVA result revealed that both the factors of culturing season and oyster size significantly affected pH, DO, $p\text{CO}_2$, and different forms of DIC, phosphates, and nitrogen in the water column of kelp-oyster polyculture mode (KO group) after 24 h of cultivation, while their interaction also significantly influenced these physicochemical parameters, except pH and CO_2 ($P < 0.05$, Table 1).

3.1.1. The pH, DO, $p\text{CO}_2$, and carbonate system

The water column of KO_S , KO_M , and KO_L groups showed a significant decrease in pH (decreased 2.02 %–3.45 %) and DO concentration (decreased 9.97 %–15.91 %), along with the $p\text{CO}_2$ significantly increased by 45.94 %–71.96 % compared to the KO_{non} group after 24 h of summer cultivation ($P < 0.05$) (Fig. 2-A, B, C). Similarly, the analysis result of the carbonate system showed that the concentrations of CO_2 , HCO_3^- , and DIC were obviously increased in the KO_S , KO_M , and KO_L groups compared to the KO_{non} group and decreased with increasing oyster size, while the CO_3^{2-} concentration was significantly decreased in these three treatment groups ($P < 0.05$) (Fig. 2-D). Among these polyculture modes, the KO_S and KO_M groups exhibited significantly lower pH and DO concentrations, along with significantly higher $p\text{CO}_2$ in the

summer compared to the KO_L group ($P < 0.05$).

The higher DO and pH levels, as well as lower $p\text{CO}_2$ during the winter culturing period, were observed in all of the experimental groups, including the KO_{non} group, compared to the summer (Fig. 2-A, B, C). Moreover, the trends of DO, pH, $p\text{CO}_2$, and the carbonate system changes in the three polyculture groups were opposite in the winter compared to the summer, compared to the KO_{non} group. Specifically, after 24 h of winter culturing the value of $p\text{CO}_2$ (decreased 6.63 %–24.23 %) and concentration of CO_2 , HCO_3^- , and DIC in the KO_S , KO_M , and KO_L groups were obviously decreased, and along with the significant increase of pH (increased 0.35 %–1.16 %) and concentration of DO (increased 6.31 %–11.30 %) and CO_3^{2-} compared to the KO_{non} group. Of which, the KO_M and KO_L groups exhibited relatively higher pH, concentrations of DO, and CO_3^{2-} , as well as relatively lower concentrations of CO_2 , HCO_3^- , and $p\text{CO}_2$ compared to the KO_S group in the winter, but with no significant differences observed between the KO_M and KO_L groups ($P > 0.05$).

3.1.2. Nutrients

The overall trends of nutrient concentration changes were obviously different across three polyculture treatments between two seasons (Fig. 3). Specifically, after 24 h culturing period, the three polyculture modes characterized by the absorption of NO_2^- -N and NO_3^- -N and the release of DIP and NH_4^+ -N in summer, while in winter, they were marked by the sharp absorption of DIP (absorbed 41.97 %–53.58 %), NO_3^- -N (absorbed 25.56 %–50.50 %), and NH_4^+ -N (absorbed 36.01 %–59.02 %), compared to the control group. Furthermore, there were significant differences in the concentrations of NO_2^- -N, NO_3^- -N, and NH_4^+ -N among the three polyculture treatments during summer, with the KO_M group exhibiting a minimum DIN concentration of 8.402 $\mu\text{mol/L}$. Significant differences were also observed in the concentrations of DIP, NO_3^- -N, and NH_4^+ -N among the three polyculture treatments during the winter culturing period, with lowest DIP concentration being 0.628 $\mu\text{mol/L}$ in the KO_M group while the lowest DIN concentration was 4.594 $\mu\text{mol/L}$ in the KO_S group. In addition, after 24 h of co-culturing, the N/P ratio of seawater in the KO_{non} group was 38.26 ± 7.65 in summer, obviously higher than the ratios of 22.03 ± 1.58 to 27.55 ± 4.18 observed in the three polyculture treatments, while in winter, the N/P ratio in the KO_{non} group was 17.32 ± 0.10 , lower than the ratios of 19.74 ± 0.87 to 20.69 ± 1.27 achieved in these three treatments.

3.2. Rates of oysters and kelp in increasing/ decreasing different forms of dissolved inorganic carbon and nutrients

3.2.1. Rates of oysters of different sizes in increasing different forms of dissolved inorganic carbon and nutrients

In the closed oyster monoculture system with sufficient DO level (more than 6.45 mg/L) (including OA_S , OA_M , and OA_L treatments) that excludes planktonic community, the concentration of DIC, DIP, and DIN in seawater increased during both summer and winter (Fig. 4), along with the pH significantly decreased in these treatments (Fig. S1-a). The contribution to changes in the carbonate system of seawater by different sizes of oysters was mainly characterized by the increase of HCO_3^- and CO_2 , along with the decrease of CO_3^{2-} . HCO_3^- was the main increasing form of DIC in the OA_S , OA_M , and OA_L treatments, with increase rates of 54.87 ± 10.20 , 35.99 ± 9.55 , and 26.69 ± 8.42 $\mu\text{mol/g/d}$ in summer, respectively, which were obviously higher than the rates of 5.09 ± 0.56 , 2.72 ± 0.35 , and 0.80 ± 0.04 $\mu\text{mol/g/d}$ in winter (Fig. 4-A). Additionally, NO_3^- -N and NH_4^+ -N as the main forms of DIN increased by oysters in the OA_S , OA_M , and OA_L treatments, with NH_4^+ -N increase rates of 2.78 ± 0.39 , 1.41 ± 0.06 , and 0.79 ± 0.05 $\mu\text{mol/g/d}$ in summer, respectively, which were obviously higher than the rates of 0.60 ± 0.06 , 0.25 ± 0.01 , and 0.15 ± 0.04 $\mu\text{mol/g/d}$ in winter (Fig. 4-B). Furthermore, the NH_4^+ -N and HCO_3^- increase rates by oysters across each treatment decreased with increasing oyster size during both summer and winter.

The correlation analysis results (Fig. 5) revealed that the

Table 1

Main results of Two-Way ANOVA for oyster size and season on physicochemical parameters of water column in kelp-oyster polyculture mode.

Environmental factors	P-value		
	Oyster size	Season	Oyster size $\times$ season
pH	< 0.01**	< 0.001***	0.447
DO (mg/L)	< 0.001***	< 0.001***	< 0.01**
$p\text{CO}_2$ (μatm)	< 0.001***	< 0.001***	< 0.01**
HCO_3^- ($\mu\text{mol/L}$)	< 0.001***	< 0.001***	< 0.01**
CO_3^{2-} ($\mu\text{mol/L}$)	< 0.001***	< 0.001***	< 0.001***
CO_2 ($\mu\text{mol/L}$)	< 0.001***	< 0.001***	0.058
DIP ($\mu\text{mol/L}$)	< 0.05*	< 0.001***	< 0.001***
NO_2^- -N ($\mu\text{mol/L}$)	< 0.05*	< 0.001***	< 0.01**
NO_3^- -N ($\mu\text{mol/L}$)	< 0.001***	< 0.001***	< 0.001***
NH_4^+ -N ($\mu\text{mol/L}$)	< 0.01**	< 0.001***	< 0.01**

Note: Asterisks represent the significant difference, * represents $P < 0.05$, ** represents $P < 0.01$, *** represents $P < 0.001$.

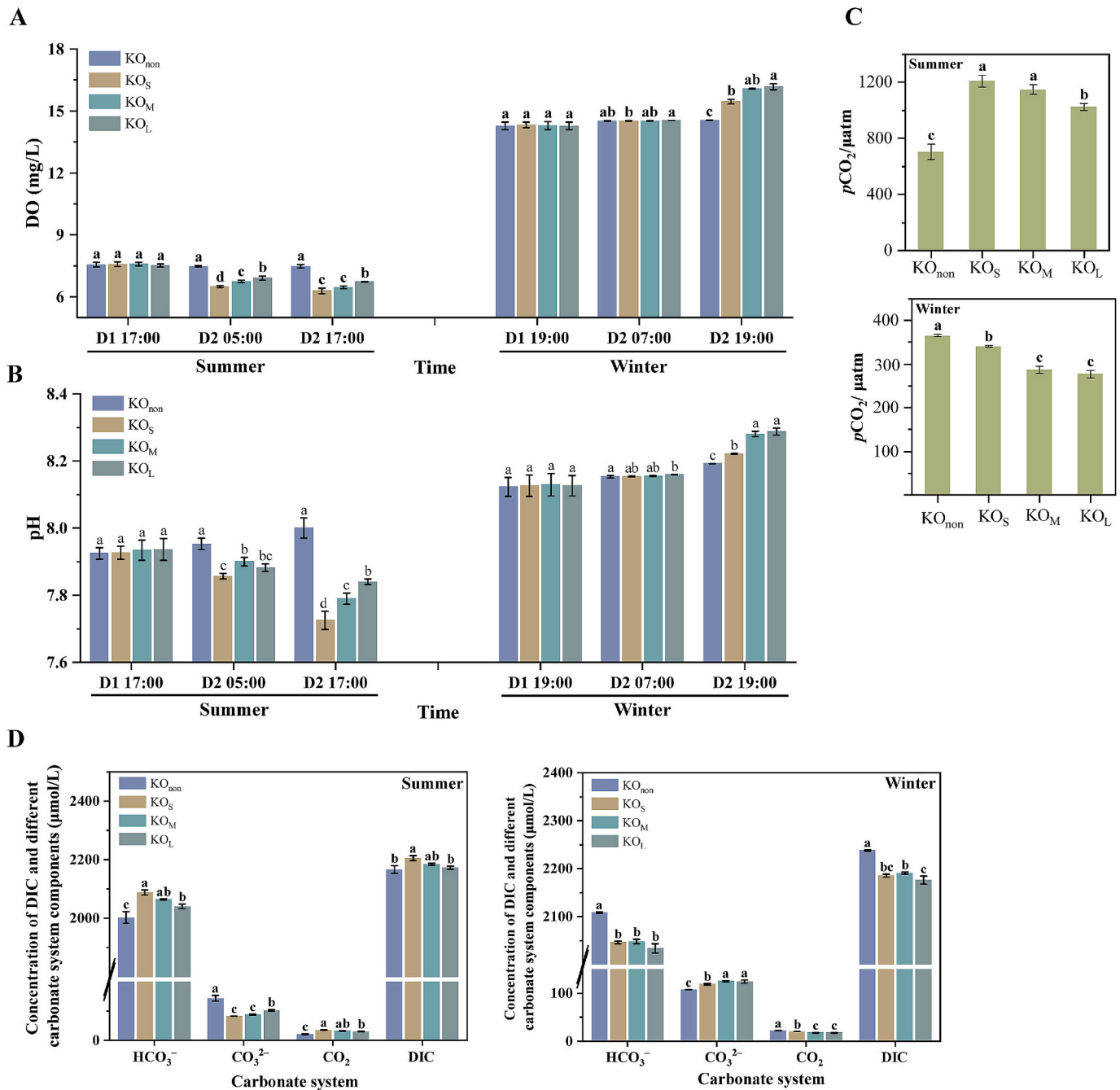


Fig. 2. The variations in (A) DO, (B) pH, (C) $p\text{CO}_2$, and (D) carbonate system in the seawater of different kelp-oyster polyculture modes and the blank control in the summer and winter after culturing 24 h. Data with different letters represent the significant differences among different groups ($P < 0.05$). D1 and D2 represent the sampling time on the first day and second day, respectively.

concentration of CO_2 exhibited a significant positive correlation with the $p\text{CO}_2$, and the concentration of HCO_3^- , $\text{PO}_4^{3-}\text{-P}$, $\text{NO}_2^-\text{-N}$, and $\text{NH}_4^+\text{-N}$ ($P < 0.05$), while showing a highly significant negative correlation with pH, and the concentration of DO and CO_3^{2-} ($P < 0.001$), in seawater of the closed oyster monoculture system. It was also demonstrated that the WD_T and WD_S of oysters both exhibited a significant negative correlation with the concentration of $\text{NO}_2^-\text{-N}$ and $\text{NH}_4^+\text{-N}$ ($P < 0.05$).

3.2.2. Rates of kelp in decreasing different forms of dissolved inorganic carbon and nutrients

In the closed kelp monoculture system that excludes planktonic community, the concentration of DIC, DIP, and DIN in seawater decreased during both summer and winter, with decrease rates of these

three parameters being significantly higher in winter than in summer ($P < 0.01$) (Fig. 6). Specifically, HCO_3^- and $\text{NH}_4^+\text{-N}$ respectively were the primary forms of DIC and DIN removed by the kelp, with HCO_3^- and $\text{NH}_4^+\text{-N}$ decrease rates of 202.12 ± 10.31 and 5.37 ± 0.67 $\mu\text{mol/g/d}$ in winter, respectively, highly significantly higher than the rates of 23.71 ± 1.72 and 1.63 ± 0.63 $\mu\text{mol/g/d}$ in summer ($P < 0.01$). Moreover, a significant increase in pH was also observed in winter ($P < 0.05$) (Fig. S1-b).

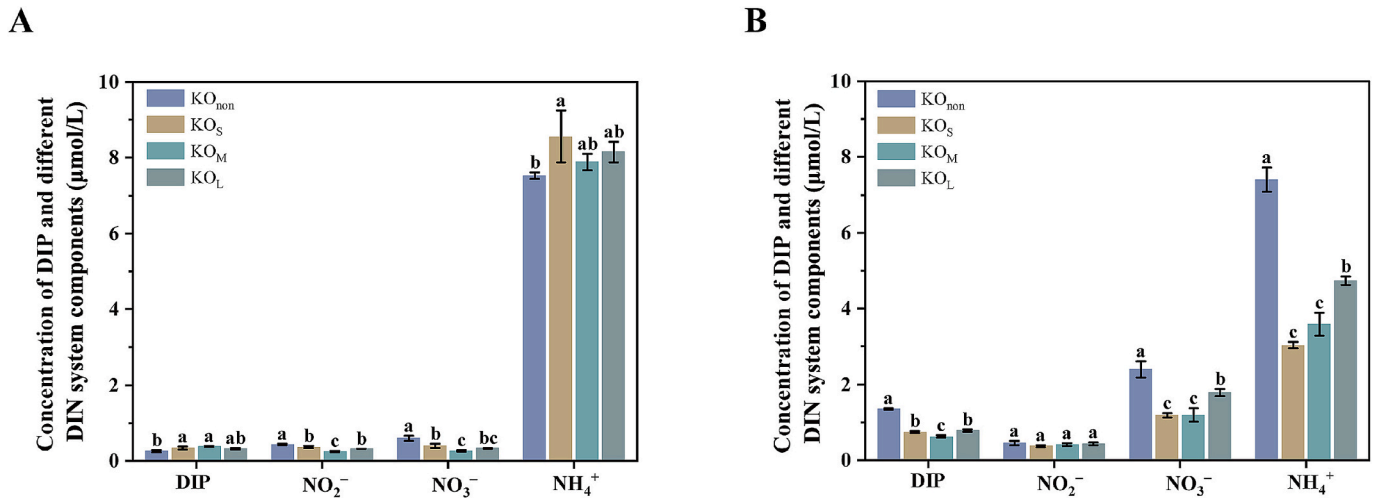


Fig. 3. The variations in DIP, NO₂⁻-N, NO₃⁻-N, and NH₄⁺-N in the seawater of different kelp-oyster polyculture modes and the blank control in the (A) summer and (B) winter after culturing 24 h. Data with different letters represent the significant differences among different groups ($P < 0.05$).

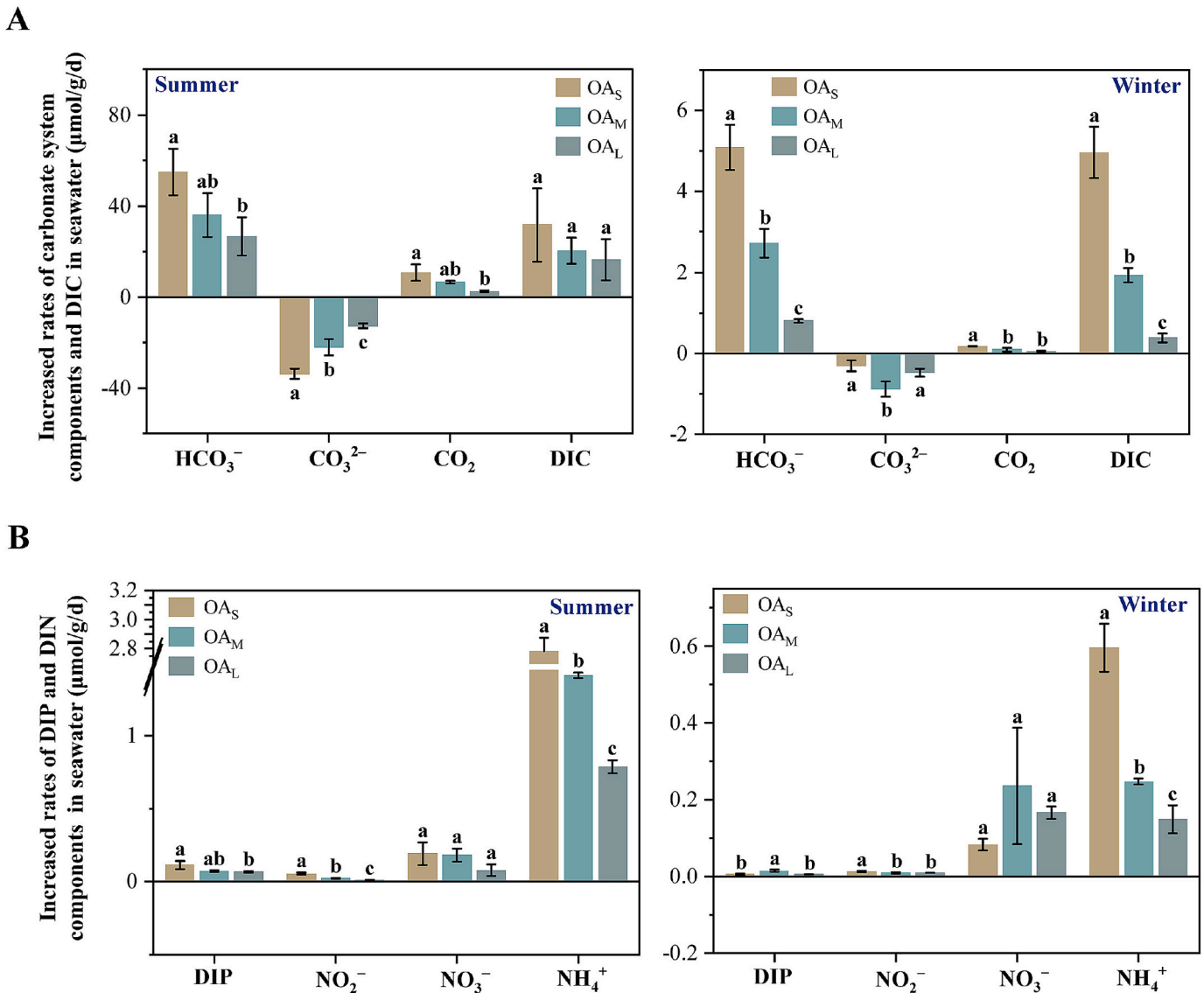


Fig. 4. The increased rates of the (A) carbonate system and (B) nutrients in seawater by different sizes of oysters in monoculture during summer and winter after 24 h of cultivation. Data with different letters represent the significant differences among different groups ($P < 0.05$).



Fig. 5. A heat map displaying the Pearson correlation coefficients of physicochemical parameters in seawater and biomass data of oysters from the closed oyster monoculture mode in the summer and winter after culturing 24 h (the correlation *P*-values were adjusted by Benjamini & Hochberg method). WD_T: dry weight of soft tissue; WD_S: dry weight of shell. Data with different asterisks represent the significant correlations among different groups (* represents *P* < 0.05, ** represents *P* < 0.01, *** represents *P* < 0.001).

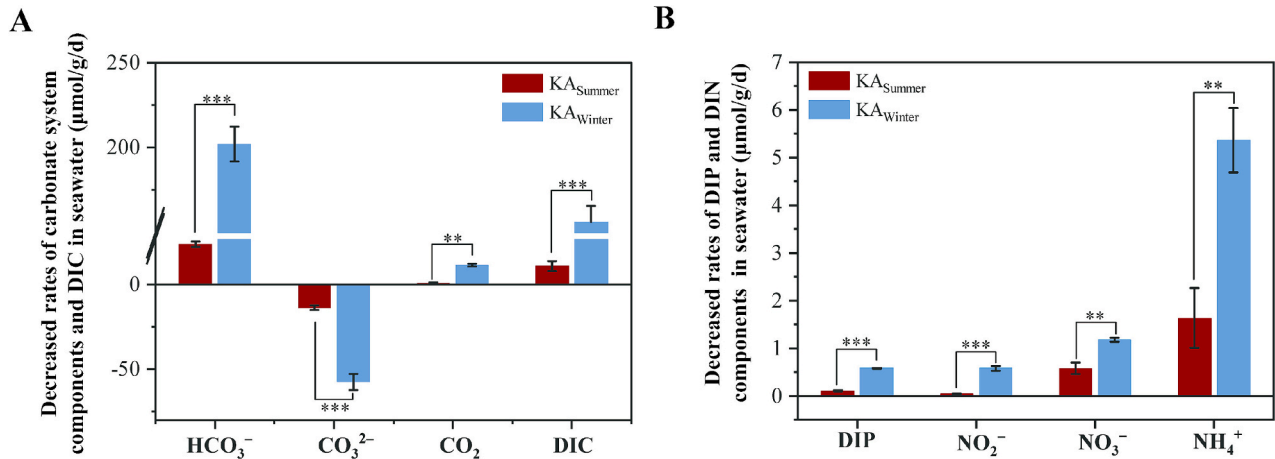


Fig. 6. The decreased rates of the (A) carbonate system and (B) nutrients in seawater by kelp in monoculture during summer (KA_{Summer}) and winter (KA_{Winter}) after 24 h of cultivation. Data with different asterisks represent the significant differences among different groups (** represents *P* < 0.01, *** represents *P* < 0.001).

3.3. Characteristics of absorbing/ releasing dissolved inorganic carbon, phosphates, and nitrogen from different components of the polyculture system

The contribution of culturing water, kelp, and oyster to the increase or decrease of CO₂, DIC, DIP, and DIN in KO_S, KO_M, and KO_L treatments

during the summer and winter after 24 h of co-culturing period was shown in Fig. 7. In these three kelp-oyster polyculture modes during summer, the CO₂ emissions from three size of oysters ranged from 1865.97 ± 364.78 μmol to 8524.03 ± 2983.48 μmol in 24 h, with the uptake of CO₂ by kelp and culturing water accounted for only 0.72 %–1.92 % and 9.24 %–20.57 % of these emissions, respectively (Fig. 7-A).

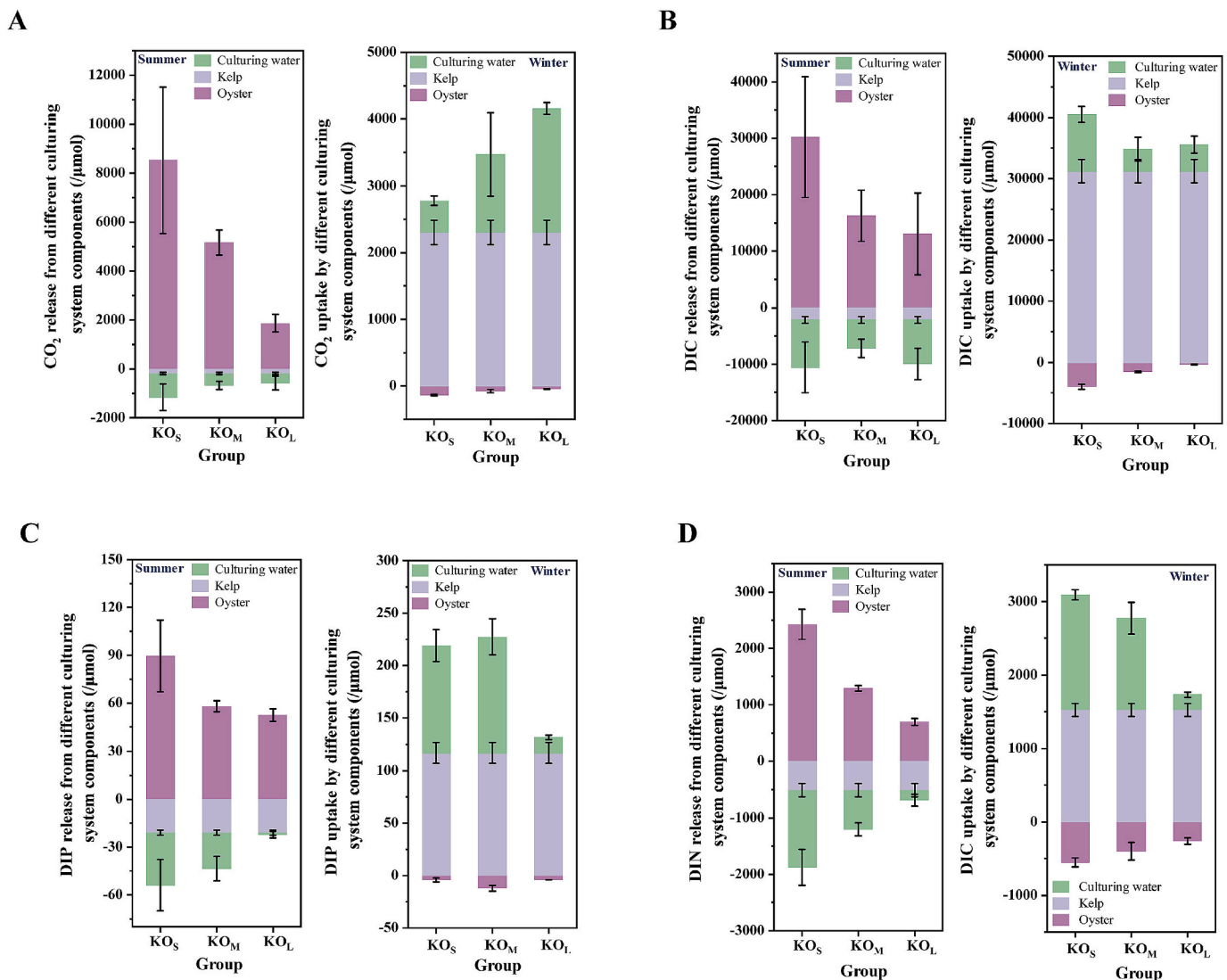


Fig. 7. The contribution of different elements in the culturing system to the increase or decrease of (A) CO₂, (B) DIC, (C) DIP, and (D) DIN in various kelp-oyster polyculture modes during summer and winter over 24 h.

On the opposite, in these kelp-oyster polyculture modes during winter in 24 h, the uptake of CO₂ by kelp and culturing water was 17.00–53.03 times and 3.49–42.70 times greater than the CO₂ emissions from three sizes of oysters, respectively.

Consistently, the combined absorption of DIC, DIP, and DIN by kelp and culturing water in these three kelp-oyster polyculture modes during summer in 24 h was found to be 10.34 %–22.49 %, 59.93 %–74.78 %, and 77.38 %–97.76 % of the release of DIC, DIP, and DIN by oysters in the respective treatments (Fig. 7-B, C, D). While, in the winter, the total absorption of DIC, DIP, and DIN by kelp and culturing water in these three kelp-oyster polyculture modes in 24 h accounted for 10.21–117.64 times, 18.85–53.24 times, and 5.54–6.85 times of the DIC, DIP, and DIN released by oysters, respectively.

4. Discussion

In this study on the field experiments of co-culturing different sizes of oysters and kelp, it was found that the physicochemical parameters of the water column in the culture system are closely linked to the metabolic activities of oysters, kelp, and the planktonic community in the cultured water under climate change. It was reported that bivalve farming activities primarily influence the carbonate system in seawater

through the respiratory metabolic activity of bivalves and their shell calcification process (Pernet et al., 2024). CO₂, mainly produced by respiration, is dissolved in seawater to form H₂CO₃, which is subsequently dissociated into H⁺, CO₃²⁻, and HCO₃⁻ increasing DIC concentrations. While the calcification involves the absorption of 2 mol HCO₃⁻ and then the release of 1 mol CO₂ and CaCO₃ precipitation, resulting in a reduction of both DIC and TA in seawater (Morris and Humphreys, 2019). In this study, it was observed that in the closed environment where the photosynthetic buffering effects of phytoplankton and macroalgae were excluded, the oyster monoculture system exhibited a significant increase in CO₂ and DIC concentrations by the end of the experiment, demonstrating that changes in DIC components are primarily influenced by respiratory processes (Li et al., 2024). In addition to influencing the DIC system in seawater, the accumulation of DIP, ammonia, and other metabolic wastes released through the metabolic excretion processes of bivalves in poorly exchanged aquatic environments may increase the risk of eutrophication, thereby adversely affecting the ecological environment. While the respiratory and excretory activities of bivalves are influenced by changes in environmental conditions and the growth status of bivalves. Consequently, in the co-culture system of kelp and oysters of different sizes during both summer and winter, it was observed that the nutrient levels and DIC in the

culturing water were significantly affected by both oyster size and seasonal factors. This pattern was found to be more pronounced in the oyster monoculture system. Previous studies have demonstrated that within an optimal temperature range (not exceeding 28 °C), rising temperatures can enhance the metabolic activity of oysters (Davis and Calabrese, 1969; Wang and Li, 2018). Accordingly, it was observed that the increase rates of CO₂, DIP, and NH₄⁺-N in seawater were significantly higher in summer (25.1 ± 0.08 °C) compared to winter (2.03 ± 0.15 °C) in different sizes of oysters monoculture system in this study. Furthermore, the increase rates of these three substances displayed a clear pattern of decreasing with increasing oyster size during the summer, which is similar to findings from other studies that demonstrated a negative correlation between oxygen consumption rates and ammonia excretion rates in certain bivalve and their dry weight of soft tissue (Feng et al., 2004; Liu et al., 2022; Zhao et al., 2011). Furthermore, given that the influence of respiration far exceeds that of calcification, although previous studies have revealed a positive correlation between oyster size and calcification rates (Waldbusser et al., 2010), the significant correlation between changes in the carbonate system of seawater in the oyster monoculture system and the dry weight of oyster shells was not observed in this study. In sum, it is essential to consider the CO₂, DIN, and DIP release characteristics of bivalves associated with different body sizes of bivalves under seasonal variations and across different cultivation periods in aquaculture practices, in order to optimize the management of integrated aquaculture systems and enhance the overall efficiency of the farming system.

Macroalgae farming can generate significant environmental benefits by playing an important role in carbon sequestration through photosynthesis utilizing inorganic carbon (HCO₃⁻ and/or CO₂) and bioremediation of eutrophic waters through nutrient uptake by depleting excess DIP, DIN in the water column (Liu et al., 2021). Therefore, it was observed that the kelp primarily exhibited the absorption of HCO₃⁻ and CO₂, where HCO₃⁻ may be the main form of inorganic carbon utilized by kelp for photosynthesis in monoculture during both summer and winter in this study (Fernández et al., 2015). Furthermore, the absorption of CO₂ and HCO₃⁻ by kelp for photosynthesis leads to obviously increased pH and CO₃²⁻ saturation in seawater of the monoculture system, which creates more favorable chemical conditions for bivalve calcification, thereby supporting the hypothesis that macroalgae farming can mitigate the adverse effects of ocean acidification on calcifying organisms (Leal et al., 2024; Narvarte et al., 2024). Moreover, as a cold-water macroalgae, relatively lower temperatures are often more conducive to the growth and photosynthesis of kelp (Sivertsen and Bjørge, 2015). Therefore, despite the enhanced metabolic activity of oysters in summer providing abundant DIC and nutrients of DIN and DIP for photosynthesis, it was still observed that kelp exhibited significantly higher efficiencies in the absorption of DIC, DIP, and DIN during winter compared to summer in this study. However, the efficiency of kelp as a “biological carbon pump” diminishes during the summer, resulting in insufficient buffering capacity to offset the carbon dioxide produced by oysters’ respiration and calcification. Consequently, the pH (decreased by 2.02 % to 3.45 %), DO concentration (decreased by 9.97 % to 15.91 %) and CO₃²⁻ saturation in the seawater of the kelp-oyster polyculture treatments significantly decreased compared to the control group, which will restrict the shell synthesis and growth of calcifying organisms (Wahl et al., 2018). Research by Liu et al. (2021) indicates that when the farming ratio of *Patinopekten yessoensis* to *Gracilaria lemaneiformis* is 1:1.6, there are no significant differences in pCO₂ and the levels of inorganic carbon system components between the polyculture treatments and the macroalgae monoculture group. Additionally, Han et al. (2013) demonstrated that appropriately increasing the proportion of macroalgae in bivalve-macroalgae polyculture systems can improve the inorganic carbon system and nutrients in the water. Therefore, it is necessary to increase the proportion of macroalgae as appropriate in integrated aquaculture systems or to cultivate heat-resistant macroalgae to mitigate the ecological pressures that large-scale bivalve farming may

impose during the summer.

The ocean can absorb approximately 30 % of the CO₂ from the atmosphere through physical and biological interactions, and the CO₂ solubility in seawater is further significantly affected by changes in environmental conditions such as water temperature, salinity, pH, etc. (Landschützer et al., 2014). Therefore, it was observed that the pCO₂ levels in the water column of the control group show typical seasonal variations during summer and winter. Additionally, planktonic communities living in the oceans that include zooplankton, phytoplankton, bacteria, etc., also play a crucial role in the biogeochemical cycles of carbon, nitrogen, and phosphorus, as well as in energy flow within marine ecosystems (Hulot et al., 2020). In contrast to the findings of Xiong et al. (2024), which observed that the respiration and remineralization of planktonic community dominated in kelp farming areas during summer (from June to August), leading to the seawater acting as a CO₂ source, this study revealed that the culturing water in different kelp-oyster polyculture modes predominantly exhibited CO₂ fixation and absorption of DIC, DIN, and DIP during both summer and winter. This result indicates that photosynthesis by photosynthetic autotrophs, such as phytoplankton and phototrophic bacteria, dominates the water column of these polyculture systems during summer and winter. As a major contributor to global marine primary productivity (accounting for over 50 %), phytoplankton are not only key participants in the marine carbon cycle but also serve as the primary food source for filter-feeding bivalves in aquaculture (Simon et al., 2009). Although, in this study, the phytoplankton community faced significant downward control effects due to intense oyster filtration during summer (Prins et al., 1995), the culturing water still demonstrated CO₂ fixation and absorption of DIC, DIN, and DIP in different polyculture modes. This phenomenon mainly be due to the oysters primarily feeding on the micro (2–20 μm) and small (20–200 μm) planktonic organisms (Du et al., 2019), which do not effectively utilize the picoplankton (0.45–2 μm) that are crucial for marine primary productivity (Glover et al., 1986). On the other hand, the DIN and DIP nutrients produced by oyster metabolism promote the growth of phytoplankton which might bring a significant increase in their biomass within 1 to 2 days (Zhang et al., 2024). These results in this study further support this hypothesis: the absorption of DIN and DIP in the culturing water of different polyculture treatments during the summer was obviously correlated with the DIN and DIP release from the corresponding size oysters, and the final N/P ratios of seawater in the polyculture treatments (22.03 ± 1.58 to 27.55 ± 4.18) were significantly lower than that of the control group (38.26 ± 7.65), indicating an alleviation of phosphate limitation.

The biomass of phytoplankton is also influenced by the upward control effects of competition for nutrients with macroalgae. When the proportion of macroalgae is low, the weak nutrient competition from macroalgae may lead to excessive proliferation of the phytoplankton community (Prins et al., 1995). In this study, it was observed that when kelp metabolic activity was heightened during winter, the cultured water showed slightly lower absorption rates of DIC, DIN, and DIP might be due to strong nutrient competition from the kelp. Moreover, consistent with the observations made in summer, the final N/P ratios in the water column of the polyculture treatments during winter (ranging from 19.74 ± 0.87 to 20.69 ± 1.27) also exceeded the Redfield ratio (N: P = 16:1) (Redfield et al., 1963). These results indicate that under the co-cultivation ratio of oysters and kelp in this study, the growth of phytoplankton was limited by inorganic phosphorus concentrations during both the summer and winter seasons. Similar observations were made by Hou et al. (2021), who found a positive correlation between phytoplankton biomass and phosphate concentration in the coastal waters of Sangou Bay as affected by aquaculture activities. At the end of the experiment, the concentrations of DIN (4.59 ± 0.14–9.32 ± 0.63 μmol/L) and PO₄³⁻-P (0.32 ± 0.02–0.79 ± 0.03 μmol/L) in the aquaculture waters under different polyculture modes met the Class I and II marine water quality standards set by the Chinese National Standard (GB 3097–1997). This suggests that under reasonable aquaculture modes,

the synergistic regulatory functions exerted by planktonic communities (particularly phytoplankton) and macroalgae provide fundamental support for the health and sustainability of IMTA system (Zhang et al., 2024).

This study analyzed the dynamic interactions among kelp, oysters, and the cultured water system under seasonal variations and different growth sizes of oysters, revealing the complex coupling mechanisms between functional units in bivalve-macroalgae co-culture systems and their regulatory effects on the physicochemical environment of the aquaculture water. These findings provide critical scientific guidance for aquaculture practices. For instance, in the management of co-culture systems, it is essential to fully consider the impacts of seasonal variations and the growth of cultured organisms on the dynamics of the inorganic carbon cycle and nutrient cycling. Specifically, during high-temperature seasons and the juvenile stages of cultured animals, enhanced water quality monitoring and regulation should be implemented to maintain environmental stability and reduce the risks of eutrophication and acidification. Simultaneously, the proportion of bivalves and macroalgae in co-culture systems should be dynamically adjusted to optimize management strategies. In summary, future aquaculture practices should adopt more refined management approaches to enhance the synergistic effects among functional units in IMTA systems, thereby maximizing production efficiency while effectively mitigating the negative environmental impacts of aquaculture activities.

5. Conclusions

The results of in situ experiments demonstrate that under the fixed cultivation ratios established in this study, the water parameters of different-sized oyster-kelp polyculture systems were significantly influenced by both oyster size and seasonal variations. The observed characteristics of releasing DIC, DIN, and DIP in summer while absorbing these components in winter by these polyculture systems provide evidence that bivalve-macroalgae polyculture systems can mitigate the negative ecological impacts associated with oyster monoculture. Additionally, in these polyculture systems, the dominance of phytoplankton photosynthesis in the cultured water remains significant, despite the filtering activity of oysters and nutrient competition from large macroalgae. This dominance compensates for the reduced ecological benefits associated with the decreased metabolic activity of macroalgae during high summer temperatures, thereby exhibiting a synergistic bioremediation effect with the macroalgae. These findings underscore the importance of considering the interactions among bivalve, macroalgae, and water columns (together constitute a system) when discussing the ecological functions, which provides suggestions for designing and optimizing bivalve-macroalgae IMTA system.

CRedit authorship contribution statement

Yumeng Xie: Writing – original draft, Visualization, Investigation, Formal analysis, Conceptualization. **Xiaofang Qin:** Investigation, Formal analysis, Data curation. **Jinghui Fang:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Jialei Xu:** Validation, Supervision, Resources. **Yitao Zhang:** Supervision, Resources. **Junwei Wang:** Supervision, Resources. **Xiaowen Zhang:** Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.aquaculture.2025.742165>.

Data availability

Data will be made available on request.

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